QC Foundations [3/4]

Formalising the Pictures

Quantum Computing — CEP

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31 October 2026

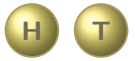
Recap — The Coin Puzzle

Trick coin



the *same* face twice

Fair coin



two *different* faces

same faces ⇒ **constant** f
different faces ⇒ **balanced** f

The four boxes $f : \{0,1\} \to \{0,1\}$

box	$f(0)$	$f(1)$	
always 0	0	0	constant
always 1	1	1	constant
copy	0	1	balanced
flip	1	0	balanced

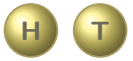
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Classical: 2 looks

One look ($f(0)$) fits one constant *and* one balanced box. You must turn the coin over.

Quantum: 1 query

Last Saturday we *watched* Deutsch do it with arrows. Today we **prove** every move.

▸ Now watch: arrows to algebra (3 min)

Picture: L2 coin puzzle; the four boxes

The Dictionary — Pictures to Symbols

Picture (L1/L2)	**Symbol (today)**	**Proved on**
Arrow on the Bloch sphere	$\|\psi\rangle = \alpha\|0\rangle + \beta\|1\rangle$	?
Shadow on the wall (measurement)	$\|\langle x\|\psi\rangle\|^2$	?
Gate = rotation that can be undone	unitary U, $U^\dagger U = I$	?
$\|-\rangle$ sits on the X axis	$X\|-\rangle = -\|-\rangle$	?
Two wires	$\otimes$	?
"Neither qubit has its own state"	cannot factor	?
The kick travels back up the wire	$(-1)^{f(x)}$	?
Arrows add or cancel	$(-1)^{f(0)} \pm (-1)^{f(1)}$	?

By the end of the lecture every "?" is replaced by a frame number.

Kets Are Column Vectors

The computational basis of $\mathbb{C}^2$

$$|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \qquad |1\rangle = \begin{pmatrix}0\\1\end{pmatrix}$$

A general qubit

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle = \begin{pmatrix}\alpha\\\beta\end{pmatrix}, \qquad \alpha,\beta \in \mathbb{C}$$

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L1's six key states

State	Dirac	Vector
$\|+\rangle$	$\frac{1}{\sqrt{2}}(\|0\rangle + \|1\rangle)$	$\frac{1}{\sqrt{2}}(1,\ 1)^T$
$\|-\rangle$	$\frac{1}{\sqrt{2}}(\|0\rangle - \|1\rangle)$	$\frac{1}{\sqrt{2}}(1,-1)^T$
$\|+i\rangle$	$\frac{1}{\sqrt{2}}(\|0\rangle + i\,\|1\rangle)$	$\frac{1}{\sqrt{2}}(1,\ i)^T$
$\|-i\rangle$	$\frac{1}{\sqrt{2}}(\|0\rangle - i\,\|1\rangle)$	$\frac{1}{\sqrt{2}}(1,-i)^T$

A ket is just a name for a column vector in linear algebra's $\mathbb{C}^2$. The label inside $|\cdot\rangle$ is whatever helps you remember it.

Picture: L1 "Ket notation"

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Bras — The Conjugate Transpose

Definition

$$\langle\psi| = |\psi\rangle^\dagger = \begin{pmatrix}\alpha^* & \beta^*\end{pmatrix}$$

A row vector: transpose *and* complex-conjugate.

Examples

$$\langle +| = \tfrac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\end{pmatrix}, \qquad \langle +i| = \tfrac{1}{\sqrt{2}}\begin{pmatrix}1 & -i\end{pmatrix}$$

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The most common mistake

Forgetting the conjugate.

$\langle +i| \neq \frac{1}{\sqrt{2}}\begin{pmatrix}1 & i\end{pmatrix}$.

For real amplitudes ($|0\rangle, |1\rangle, |\pm\rangle$) the conjugate does nothing, which is exactly why the mistake goes unnoticed until i shows up.

Rule of thumb: a ket goes in, a bra measures against it.

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The Bracket — Inner Product

Bra times ket

$$\langle\varphi|\psi\rangle = \langle\varphi|\,|\psi\rangle = \varphi_0^*\psi_0 + \varphi_1^*\psi_1 \in \mathbb{C}$$

Properties: $\langle\psi|\psi\rangle = \|\psi\|^2 \geq 0$;
$\langle\varphi|\psi\rangle = \langle\psi|\varphi\rangle^*$.

Orthonormality

$$\langle 0|0\rangle = \langle 1|1\rangle = 1, \qquad \langle 0|1\rangle = 0$$
$$\langle +|+\rangle = 1, \qquad \langle +|-\rangle = \tfrac{1}{2}(1-1) = 0$$

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Worked: $\langle +|+i\rangle$

$$\langle +|+i\rangle = \tfrac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\end{pmatrix}\tfrac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix} = \frac{1+i}{2}$$

$$|\langle +|+i\rangle|^2 = \frac{|1+i|^2}{4} = \frac{2}{4} = \frac{1}{2}$$

Geometric meaning: how much of $|\psi\rangle$ lies along $|\varphi\rangle$. $|+\rangle$ and $|+i\rangle$ are 90° apart on the sphere, so the overlap is $\frac{1}{2}$.

The Born Rule, Formally

Postulate

Measuring $|\psi\rangle$ in an orthonormal basis $\{|x\rangle\}$ gives outcome x with probability

$$P(x) = |\langle x|\psi\rangle|^2,$$

and the state afterwards is $|x\rangle$.

Normalisation

$$\sum_x P(x) = |\alpha|^2 + |\beta|^2 = \langle\psi|\psi\rangle = 1$$

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Worked: $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$

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Worked: $|+i\rangle$

$$P(0) = |\langle 0|+i\rangle|^2 = \tfrac{1}{2}$$

$$P(1) = |\langle 1|+i\rangle|^2 = \left|\tfrac{i}{\sqrt{2}}\right|^2 = \tfrac{1}{2}$$

Same as $|+\rangle$ and $|-\rangle$: the Z basis cannot tell them apart.

Picture: L1 shadow analogy; "latitude controls probability"

Measuring in Any Basis

Any orthonormal pair works as a measurement. The X basis:

$$P(+) = |\langle +|\psi\rangle|^2, \qquad P(-) = |\langle -|\psi\rangle|^2$$

Worked: same $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$

$$\langle +|\psi\rangle = \frac{\sqrt{3}+1}{2\sqrt{2}} \;\Rightarrow\; P(+) = \frac{2+\sqrt{3}}{4} \approx 0.93$$

$$P(-) = \frac{2-\sqrt{3}}{4} \approx 0.07$$

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Three states, two bases

State	$P(0)$	$P(+)$
$\|+\rangle$	$\frac{1}{2}$	1
$\|-\rangle$	$\frac{1}{2}$	0
$\|+i\rangle$	$\frac{1}{2}$	$\frac{1}{2}$

Identical in the Z basis; completely different in the X basis.

In a circuit: apply H, then measure in Z. That is exactly what the last H in Deutsch's circuit does.

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Picture: L1 "relative phase is observable"; L2 "decode" step

Global vs Relative Phase

Global phase is invisible

For any $\gamma \in \mathbb{R}$ and any basis state $|x\rangle$:

$$|\langle x|e^{i\gamma}\psi\rangle|^2 = |e^{i\gamma}|^2\,|\langle x|\psi\rangle|^2 = |\langle x|\psi\rangle|^2$$

No measurement in any basis can detect it, so $|\psi\rangle$ and $e^{i\gamma}|\psi\rangle$ are the same physical state.

Special case $\gamma = \pi$: $-|\psi\rangle \equiv |\psi\rangle$.

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Relative phase is real

$$|+\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle+|1\rangle), \qquad |-\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

The sign sits on *one* term, not on the whole state. Frame 8: $P(+) = 1$ versus $P(+) = 0$.

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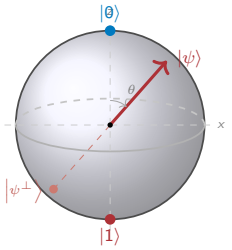
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Payoff for L2

The "always 1" oracle multiplied the *whole* state by -1, which is invisible. The balanced oracles put -1 on *one* branch, which is a relative phase and therefore visible.

The Bloch Parametrisation



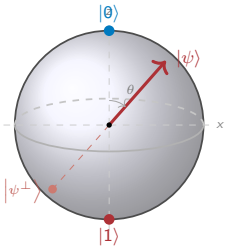
Picture: L1 latitude = probability, longitude = phase

Every state, up to global phase

$$|\psi\rangle = \cos\tfrac{\theta}{2}\,|0\rangle + e^{i\varphi}\sin\tfrac{\theta}{2}\,|1\rangle\,, \quad \theta \in [0,\pi],\ \varphi \in [0,2\pi)$$

Global phase makes α real; normalisation leaves two angles.

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Why $\theta/2$? Opposite points are orthogonal

$|\psi^{\perp}\rangle$ has $\theta \to \pi - \theta$, $\varphi \to \varphi + \pi$:

$$\left\langle \psi \middle| \psi^{\perp} \right\rangle = \cos\tfrac{\theta}{2}\sin\tfrac{\theta}{2} - \sin\tfrac{\theta}{2}\cos\tfrac{\theta}{2} = 0$$

180° on the sphere is 90° in $\mathbb{C}^2$.

Poll — Orthogonal or Not?

Question

Are $|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$ and $|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle)$ orthogonal?

A) Yes **B)** No, overlap 1 **C)** No, overlap $\frac{1}{2}$

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A) Yes **B)** No, overlap 1 **C)** No, overlap $\frac{1}{2}$

Correct (with the conjugate)

$$\langle +i|-i\rangle = \tfrac{1}{2}(1\cdot 1 + (-i)(-i)) = \tfrac{1}{2}(1-1) = 0$$

A — orthogonal: opposite ends of the y axis.

Wrong (forgot the conjugate)

$$\tfrac{1}{2}(1\cdot 1 + (i)(-i)) = \tfrac{1}{2}(1+1) = 1$$

This would say they are the *same* state, which is nonsense.

Outer Products Are Operators

Ket times bra = a matrix

$$|u\rangle\langle v| = \begin{pmatrix} u_0 \\ u_1 \end{pmatrix}\begin{pmatrix} v_0^* & v_1^* \end{pmatrix} = \begin{pmatrix} u_0 v_0^* & u_0 v_1^* \\ u_1 v_0^* & u_1 v_1^* \end{pmatrix}$$

How it acts

$$\big(\,|u\rangle\langle v|\,\big)\,|w\rangle = |u\rangle\,\langle v|w\rangle$$

"Measure how much the input looks like v; output that much u."

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"Measure how much the input looks like v; output that much u."

Worked: $|1\rangle\langle 0|$

$$|1\rangle\langle 0| = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$|1\rangle\langle 0| \; |0\rangle = |1\rangle \, \langle 0|0\rangle = |1\rangle$

$|1\rangle\langle 0| \; |1\rangle = |1\rangle \, \langle 0|1\rangle = 0$

$|i\rangle\langle j|$ is the matrix with a single 1 in row i, column j. Every 2×2 matrix is a sum of four of them.

Completeness and Projectors

Completeness (resolution of the identity)

For any orthonormal basis:

$$|0\rangle\langle 0| + |1\rangle\langle 1| = I = |+\rangle\langle +| + |-\rangle\langle -|$$

Proof: apply to any $|\psi\rangle$: you get
$|0\rangle \, \langle 0|\psi\rangle + |1\rangle \, \langle 1|\psi\rangle = \alpha \, |0\rangle + \beta \, |1\rangle$.

Use: insert I anywhere to change basis.

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Projectors

$P_x = |x\rangle\langle x|$ keeps the part of a state along $|x\rangle$:

$$P_x^2 = P_x, \qquad P_x \, |\psi\rangle = \langle x|\psi\rangle \; |x\rangle$$

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Born rule, again

$$P(x) = |\, \langle x|\psi\rangle \,|^2 = \langle \psi| \, P_x \, |\psi\rangle$$

Picture: L1 shadow = projection onto an axis

Gates as Outer Products

Gate	In words	Dirac	Matrix
X	swap $0 \leftrightarrow 1$	$\|1\rangle\langle 0\| + \|0\rangle\langle 1\|$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Z	sign on $\|1\rangle$	$\|0\rangle\langle 0\| - \|1\rangle\langle 1\|$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
H	$0 \to +,\ 1 \to -$	$\|+\rangle\langle 0\| + \|-\rangle\langle 1\|$	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

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Recipe

A gate that sends $|j\rangle$ to $|u_j\rangle$ is

$$U = \sum_j |u_j\rangle\langle j|$$

Read the Dirac form as a list of instructions.

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Check H: $|+\rangle\langle 0| + |-\rangle\langle 1| = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} + \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Picture: L1 X, H, Z frames

Checking Them — Algebra in One Line

$X|0\rangle$

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$H|+\rangle$ — L1's "H twice"

H is its own conjugate transpose ($H^\dagger = H$), so also $H = |0\rangle\langle +| + |1\rangle\langle -|$. Then

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Compare L1's version: expand, collect, watch $|1\rangle$ cancel. Here the cancellation is the single fact $\langle -|+\rangle = 0$.

Picture: L1 "Apply H twice — interference"

Unitarity — Why Gates Can Be Undone

Definition

U is unitary if $U^\dagger U = UU^\dagger = I$, i.e. $U^{-1} = U^\dagger$.

It preserves every inner product

If $|\psi'\rangle = U|\psi\rangle$ and $|\varphi'\rangle = U|\varphi\rangle$:

$$\langle\varphi'|\psi'\rangle = \langle\varphi|\,U^\dagger U\,|\psi\rangle = \langle\varphi|\psi\rangle$$

So lengths stay 1 (probabilities still sum to 1), and angles are kept: it is a rotation.

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So lengths stay 1 (probabilities still sum to 1), and angles are kept: it is a rotation.

Worked: $H^\dagger H = I$

$$\begin{aligned} H^\dagger H &= \big(|0\rangle\langle+| + |1\rangle\langle-|\big) \\ &\quad\cdot\big(|+\rangle\langle 0| + |-\rangle\langle 1|\big) \\ &= |0\rangle\langle 0|\,\langle+|+\rangle + |1\rangle\langle 1|\,\langle-|-\rangle \\ &\quad + \underbrace{|0\rangle\langle 1|\,\langle+|-\rangle + |1\rangle\langle 0|\,\langle-|+\rangle}_{0} \\ &= I \end{aligned}$$

Picture: L2 "gates are rotations; every rotation can be undone"

Eigenvectors — What a Gate Leaves Alone

Gate	Eigenvector	Eigenvalue
Z	$\|0\rangle$	$+1$
	$\|1\rangle$	-1
X	$\|+\rangle$	$+1$
	$\|-\rangle$	$-\mathbf{1}$

Spectral form (Alinear algebra, Maths [2/3])

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1|$$
$$X = |+\rangle\langle +| - |-\rangle\langle -|$$

A gate is fixed by its eigenvectors and their eigenvalues.

The key line for today

$$\begin{aligned} X|-\rangle &= \tfrac{1}{\sqrt{2}}\big(X|0\rangle - X|1\rangle\big) \\ &= \tfrac{1}{\sqrt{2}}\big(|1\rangle - |0\rangle\big) = -|-\rangle \end{aligned}$$

On the sphere: the eigenvectors of a rotation are the two points on its axis. X is a half-turn about the x axis, so $|\pm\rangle$ stay where they are and pick up only a factor (± 1).

Picture: L2 "$|-\rangle$ sits on the X axis"

Eigenvectors — What a Gate Leaves Alone

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	$\|1\rangle$	-1
X	$\|+\rangle$	$+1$
	$\|-\rangle$	$-\mathbf{1}$

$HZH = X$, Proved

Three lines

$$\begin{aligned} HZH &= H\big(|0\rangle\langle 0| - |1\rangle\langle 1|\big)H \\ &= \big(H|0\rangle\big)\big(\langle 0|H\big) - \big(H|1\rangle\big)\big(\langle 1|H\big) \\ &= |+\rangle\langle +| - |-\rangle\langle -| \;=\; X \end{aligned}$$

using $H|0\rangle = |+\rangle$, $\langle 0|H = (H|0\rangle)^\dagger = \langle +|$, and frame 17.

$HZH = X$, Proved

Three lines

$$\begin{aligned} HZH &= H\big(\,|0\rangle\langle 0| - |1\rangle\langle 1|\,\big)H \\ &= \big(H\,|0\rangle\,\big)\big(\,\langle 0|\,H\big) - \big(H\,|1\rangle\,\big)\big(\,\langle 1|\,H\big) \\ &= |+\rangle\langle +| - |-\rangle\langle -| \;=\; X \end{aligned}$$

using $H\,|0\rangle = |+\rangle$, $\langle 0|\,H = (H\,|0\rangle)^\dagger = \langle +|$, and frame 17.

The general rule

$$\begin{aligned} U\Big(\sum_k \lambda_k\,|v_k\rangle\langle v_k|\,\Big)U^\dagger \\ = \sum_k \lambda_k\,|Uv_k\rangle\langle Uv_k| \end{aligned}$$

Conjugating by U moves the eigenvectors and keeps the eigenvalues. H swaps the z and x axes, so it turns Z into X.

Gates & Circuit Design (this afternoon) builds a whole toolbox of identities like this one.

Picture: L1 poll "$HZH\,|0\rangle$"; L2 sandbox Box C

Poll — What Is HXH?

Question

Using the method of the last frame, $HXH = ?$

A) X **B)** Z **C)** I **D)** $-X$

Poll — What Is HXH?

Question

Using the method of the last frame, $HXH = ?$

A) X **B)** Z **C)** I **D)** $-X$

Answer: B

$$HXH = H\big(\,|+\rangle\langle +| - |-\rangle\langle -|\,\big)H = |0\rangle\langle 0| - |1\rangle\langle 1| = Z$$

since $H\,|+\rangle = |0\rangle$ and $H\,|-\rangle = |1\rangle$.

Circuit meaning: a Z sandwiched between Hs is an X, and vice versa. A CNOT sandwiched between Hs on the target becomes a controlled-Z.

Break

10 minutes

Tensor Products — linear algebra Recap

Two qubits live in $\mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$

$$|a\rangle \otimes |b\rangle = |a\rangle\,|b\rangle = |ab\rangle$$

$$\begin{pmatrix} a_0 \\ a_1 \end{pmatrix} \otimes \begin{pmatrix} b_0 \\ b_1 \end{pmatrix} = \begin{pmatrix} a_0 b_0 \\ a_0 b_1 \\ a_1 b_0 \\ a_1 b_1 \end{pmatrix}$$

Example: $|0\rangle \otimes |1\rangle = (0,1,0,0)^T = |01\rangle$

The four basis states

Ket	Vector
$\|00\rangle$	$e_1 = (1,0,0,0)^T$
$\|01\rangle$	$e_2 = (0,1,0,0)^T$
$\|10\rangle$	$e_3 = (0,0,1,0)^T$
$\|11\rangle$	$e_4 = (0,0,0,1)^T$

Convention today: the left label is the top wire. Qiskit prints bit strings the other way round ($q_1 q_0$).

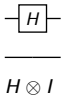
Picture: L1 "Two qubits — the 4-state space"

Gates on Wires — the Kronecker Rule

Gates act wire by wire

$$(A \otimes B)\big(|a\rangle \otimes |b\rangle\big) = A|a\rangle \otimes B|b\rangle$$

"H on the top wire only" is $H \otimes I$.



$H \otimes I$

Picture: L2 "quantum parallelism"

Gates on Wires — the Kronecker Rule

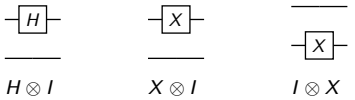
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As matrices: each entry of A scales B

$$A \otimes B = \begin{pmatrix} a_{00}B & a_{01}B \\ a_{10}B & a_{11}B \end{pmatrix} \qquad (4 \times 4)$$



$H \otimes I$ $\quad X \otimes I$ $\quad I \otimes X$

Order matters

$$X \otimes I = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \qquad I \otimes X = \begin{pmatrix} X & 0 \\ 0 & X \end{pmatrix}$$

Flip the top wire vs flip the bottom wire.

Picture: L2 "quantum parallelism"

Gates on Wires — the Kronecker Rule

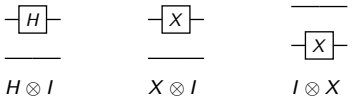
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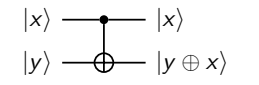
Flip the top wire vs flip the bottom wire.

Linearity does the rest

Each term separately: $U\big(\sum_x c_x |x\rangle\big) = \sum_x c_x\, U|x\rangle$, L2's "both lanes in one pass".

Picture: L2 "quantum parallelism"

CNOT, Built from Tensors



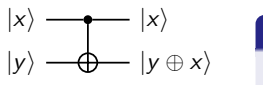
$\oplus$ = addition mod 2 (XOR)

$P_0 = |0\rangle\langle 0|, \;\; P_1 = |1\rangle\langle 1|$

"If the control is 1, apply X", in Dirac

$$\text{CNOT} = P_0 \otimes I + P_1 \otimes X$$

CNOT, Built from Tensors



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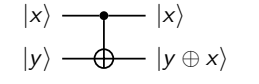
"If the control is 1, apply X", in Dirac

$$\text{CNOT} = P_0 \otimes I + P_1 \otimes X$$

$$\underbrace{\begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix}}_{P_0 \otimes I} + \underbrace{\begin{pmatrix} 0 & 0 \\ 0 & X \end{pmatrix}}_{P_1 \otimes X} = \begin{pmatrix} I & 0 \\ 0 & X \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Column $|10\rangle$ has its 1 in row $|11\rangle$: $\text{CNOT}\,|10\rangle = |11\rangle$.

CNOT, Built from Tensors



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$I \otimes P_0 + X \otimes P_1$

$$\underbrace{\begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix}}_{P_0 \otimes I} + \underbrace{\begin{pmatrix} 0 & 0 \\ 0 & X \end{pmatrix}}_{P_1 \otimes X} = \begin{pmatrix} I & 0 \\ 0 & X \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

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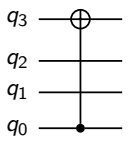
Control on the bottom wire instead? Same recipe, factors swap places

$$I \otimes P_0 + X \otimes P_1 = \begin{pmatrix} P_0 & P_1 \\ P_1 & P_0 \end{pmatrix} \qquad \text{swaps } |01\rangle \leftrightarrow |11\rangle, \text{ leaves } |00\rangle, |10\rangle$$

The order of the factors is the order of the wires.

Picture: L1 CNOT truth table

n Qubits — Same Recipe



CNOT(q_0, q_3): control q_0, target q_3

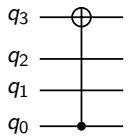
$|x_3 x_2 x_1 x_0\rangle \mapsto |(x_3 \oplus x_0)\, x_2 x_1 x_0\rangle$

Recipe: one factor per wire, top wire first

Control $\to P_0$ / P_1; target $\to I$ / X; idle $\to I$.

$$\text{CNOT}(q_0, q_3) = \underset{q_3}{I} \otimes \underset{q_2}{I} \otimes \underset{q_1}{I} \otimes \underset{q_0}{P_0} + \underset{q_3}{X} \otimes \underset{q_2}{I} \otimes \underset{q_1}{I} \otimes \underset{q_0}{P_1}$$

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As a 16×16 matrix: Kronecker rule on the first factor

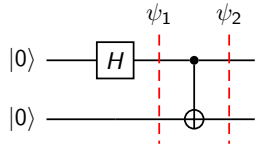
$$= \begin{pmatrix} I_4 \otimes P_0 & I_4 \otimes P_1 \\ I_4 \otimes P_1 & I_4 \otimes P_0 \end{pmatrix} \qquad (8 \times 8 \text{ blocks})$$

Even rows ($q_0 = 0$) stay put; each odd row swaps with the row 8 away.

n qubits: 2^n amplitudes and a $2^n \times 2^n$ matrix, but the recipe is always n factors long.

All 256 entries

Deriving the Bell State

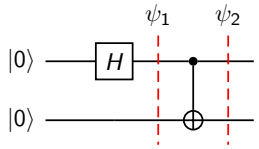


$|\psi_0\rangle = |00\rangle$

$|\psi_1\rangle = (H \otimes I)\,|00\rangle = \frac{1}{\sqrt{2}}\big(\,|00\rangle + |10\rangle\,\big)$

$|\psi_2\rangle = \mathrm{CNOT}\;|\psi_1\rangle = \frac{1}{\sqrt{2}}\big(\mathrm{CNOT}\,|00\rangle + \mathrm{CNOT}\,|10\rangle\,\big) = \frac{1}{\sqrt{2}}\big(\,|00\rangle + |11\rangle\,\big) = \big|\Phi^+\big\rangle$

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All four Bell states

$|\Phi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \quad |\Psi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$

Start from $|ab\rangle$ instead of $|00\rangle$ to get the other three (problem set).

Picture: L1 "Building a Bell pair"

Entangled Means "Cannot Factor"

Definition

A two-qubit state is a **product state** if it equals $|a\rangle \otimes |b\rangle$ for some single-qubit states. Otherwise it is **entangled**.

A product always looks like this

$$(a\,|0\rangle + b\,|1\rangle)(c\,|0\rangle + d\,|1\rangle)$$
$$= ac\,|00\rangle + ad\,|01\rangle + bc\,|10\rangle + bd\,|11\rangle$$

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$$= ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$$

Proof that $|\Phi^+\rangle$ is entangled

Matching coefficients with $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$:

$$ac = \tfrac{1}{\sqrt{2}}, \quad bd = \tfrac{1}{\sqrt{2}}, \quad ad = 0, \quad bc = 0$$

$ad = 0$ means $a = 0$ or $d = 0$.
$a = 0$ contradicts $ac \neq 0$; $d = 0$ contradicts $bd \neq 0$. ■

Quick test: $c_{00}c_{11} = c_{01}c_{10}$ holds exactly for product states.

Picture: L1 "neither qubit has a definite state on its own"

Measuring One Qubit of a Pair

Rule

Measure the first qubit and get x:

$$P(x) = \left\| (|x\rangle\langle x| \otimes I)\,|\psi\rangle \right\|^2,$$

and the state becomes that projection, renormalised.

Worked: $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

$(|0\rangle\langle 0| \otimes I)\,|\Phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle$

$P(0) = \frac{1}{2}$; the state after is $|00\rangle$.

Likewise $P(1) = \frac{1}{2}$, leaving $|11\rangle$.

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Likewise $P(1) = \frac{1}{2}$, leaving $|11\rangle$.

Perfect correlation

Whatever the first qubit shows, the second is then certain to show the same.

Each qubit on its own is a 50/50 coin, so no message is sent. Why this cannot be used to signal (and what Bob's qubit "looks like" before he is told) comes with teleportation on 7 November.

Picture: L1 "spooky correlations"

The Oracle U_f

Definition

$$U_f\,|x\rangle\,|y\rangle = |x\rangle\,|y \oplus f(x)\rangle$$

"Keep x; flip the helper if $f(x) = 1$."

It is unitary

$U_f U_f\,|x\rangle\,|y\rangle = |x\rangle\,|y \oplus f(x) \oplus f(x)\rangle = |x\rangle\,|y\rangle$, so $U_f^{-1} = U_f$. It permutes basis states, so it is a permutation matrix, which is unitary.

L2's four boxes as matrices

always 0	$I \otimes I$
always 1	$I \otimes X$
copy	CNOT
flip	$(I \otimes X)\,\mathrm{CNOT}$

$$U_{\text{flip}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Picture: L2 "making the box quantum"; "the four oracles"

The Phase Kickback Lemma

Lemma

$$U_f |x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle$$

Proof

$$U_f |x\rangle |-\rangle = |x\rangle \tfrac{1}{\sqrt{2}}\Big(|0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle \Big)$$

$f(x) = 0$: $|x\rangle \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |x\rangle |-\rangle$

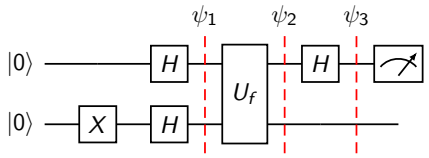
$f(x) = 1$: $|x\rangle \frac{1}{\sqrt{2}}(|1\rangle - |0\rangle) = -|x\rangle |-\rangle$ ■

Reading it

- The helper ends as $|-\rangle$ either way: it does not move
- The answer $f(x)$ becomes a sign on the input branch
- It is frame 17 again: $|-\rangle$ is the -1 eigenvector of X

Picture: L2 kickback poll: "the control moves"

Deutsch's Algorithm, Line by Line



$$|\psi_0\rangle = |0\rangle |1\rangle$$
$$|\psi_1\rangle = |+\rangle |-\rangle = \tfrac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle \big) |-\rangle$$
$$|\psi_2\rangle = \tfrac{1}{\sqrt{2}}\Big((-1)^{f(0)} |0\rangle + (-1)^{f(1)} |1\rangle \Big) |-\rangle$$
$$|\psi_3\rangle = \tfrac{1}{2}\Big[\big((-1)^{f(0)} + (-1)^{f(1)}\big) |0\rangle + \big((-1)^{f(0)} - (-1)^{f(1)}\big) |1\rangle \Big] |-\rangle$$

ψ_1: $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$. ψ_2: linearity, then the lemma (frame 29) on each term. ψ_3: H on the top qubit, collect terms.

Picture: L2 blueprint and step-by-step frames

Constant vs Balanced — The Arrows Were Coefficients

Constant: $f(0) = f(1)$

$$(-1)^{f(0)} + (-1)^{f(1)} = \pm 2$$
$$(-1)^{f(0)} - (-1)^{f(1)} = 0$$

$|\psi_3\rangle = \pm |0\rangle |-\rangle$, so it reads **0** with probability 1.

Balanced: $f(0) \neq f(1)$

$$(-1)^{f(0)} + (-1)^{f(1)} = 0$$
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$|\psi_3\rangle = \pm |1\rangle |-\rangle$, so it reads **1** with probability 1.

L2's arrows, decoded

The coefficient of $|0\rangle$ is a *sum of two route amplitudes*: one through lane 0, one through lane 1.

Same signs ⇒ they add (constructive).
Opposite signs ⇒ they cancel (destructive).

The leftover $\pm$

It multiplies the whole state, so it is a global phase (frame 9) and cannot be seen. This was L2's "always 1" poll.

Hadamard on n Qubits

One qubit, one formula

$$H|x\rangle = \tfrac{1}{\sqrt{2}}\big(|0\rangle + (-1)^x|1\rangle\big) = \tfrac{1}{\sqrt{2}}\sum_{z\in\{0,1\}}(-1)^{xz}|z\rangle$$

Check: $x=0$ gives $|+\rangle$; $x=1$ gives $|-\rangle$.

n qubits: multiply out

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}}\sum_{z\in\{0,1\}^n}(-1)^{x\cdot z}|z\rangle$$

where $x\cdot z = x_1z_1 \oplus \cdots \oplus x_nz_n$.

Special case: all zeros

$$H^{\otimes n}|0^n\rangle = \frac{1}{\sqrt{2^n}}\sum_z |z\rangle$$

The equal superposition of all 2^n inputs.

$n=2$, $x=10$

$$H^{\otimes 2}|10\rangle = \tfrac{1}{2}\big(|00\rangle + |01\rangle - |10\rangle - |11\rangle\big)$$

$= |-\rangle|+\rangle$: the sign is -1 exactly when $z_1 = 1$.

Deutsch–Jozsa for n Bits

$$|0^n\rangle|1\rangle \xrightarrow{H^{\otimes n}\otimes H} \frac{1}{\sqrt{2^n}}\sum_x|x\rangle|-\rangle \xrightarrow{U_f} \frac{1}{\sqrt{2^n}}\sum_x(-1)^{f(x)}|x\rangle|-\rangle$$
$$\xrightarrow{H^{\otimes n}\otimes I} \sum_z\left[\frac{1}{2^n}\sum_x(-1)^{f(x)+x\cdot z}\right]|z\rangle|-\rangle$$

Deutsch–Jozsa for n Bits

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$$\xrightarrow{H^{\otimes n}\otimes I} \sum_z\left[\frac{1}{2^n}\sum_x(-1)^{f(x)+x\cdot z}\right]|z\rangle|-\rangle$$

Look only at $z = 0^n$

$$\text{amplitude}(0^n) = \frac{1}{2^n}\sum_x(-1)^{f(x)}$$

Constant: all 2^n terms equal, so ± 1: always reads 0^n.
Balanced: half $+1$, half -1, so 0: never reads 0^n.

This is L2's "2^n routes" picture: each x is a route to $|0^n\rangle$ with arrow $(-1)^{f(x)}$. Constant: all arrows aligned. Balanced: they pair off and cancel exactly.

One query to U_f, for every n.

Picture: L2 "from 1 bit to n bits"

The Dictionary — Completed

Picture (L1/L2)	Symbol (today)	Proved on
Arrow on the Bloch sphere	$\|\psi\rangle = \alpha\|0\rangle + \beta\|1\rangle$	frame 10
Shadow on the wall (measurement)	$\|\langle x\|\psi\rangle\|^2$	frame 7
Gate = rotation that can be undone	unitary U, $U^\dagger U = I$	frame 16
$\|-\rangle$ sits on the X axis	$X\|-\rangle = -\|-\rangle$	frame 17
Two wires	$\otimes$	frame 21
"Neither qubit has its own state"	cannot factor	frame 26
The kick travels back up the wire	$(-1)^{f(x)}$	frame 29
Arrows add or cancel	$(-1)^{f(0)} \pm (-1)^{f(1)}$	frame 30

Pictures for intuition, algebra for proof. You now have both for everything so far.

Formula Card

States and measurement

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, $\langle\psi| = |\psi\rangle^\dagger$

$\langle\varphi|\psi\rangle = \sum_i \varphi_i^* \psi_i$

$P(x) = |\langle x|\psi\rangle|^2 = \langle\psi| P_x |\psi\rangle$

$e^{i\gamma}|\psi\rangle \equiv |\psi\rangle$ (global phase)

$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle$

Many qubits

$(A \otimes B)(|a\rangle|b\rangle) = A|a\rangle \otimes B|b\rangle$

$A \otimes B = \begin{pmatrix} a_{00}B & a_{01}B \\ a_{10}B & a_{11}B \end{pmatrix}$

$\mathrm{CNOT} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X$

C-U: $P_{0/1}$ on control, I/U on target, I elsewhere

$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)$, $|\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$

Product $\iff c_{00}c_{11} = c_{01}c_{10}$

Operators

$(|u\rangle\langle v|)|w\rangle = \langle v|w\rangle|u\rangle$; $\sum_x |x\rangle\langle x| = I$

$X = |0\rangle\langle 1| + |1\rangle\langle 0| = |+\rangle\langle +| - |-\rangle\langle -|$

$Z = |0\rangle\langle 0| - |1\rangle\langle 1|$; $H = |+\rangle\langle 0| + |-\rangle\langle 1|$

$U^\dagger U = I$; $HZH = X$, $HXH = Z$

Algorithms

$U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$

$U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$

$H^{\otimes n}|x\rangle = 2^{-n/2}\sum_z (-1)^{x\cdot z}|z\rangle$

$\mathrm{amp}(0^n) = 2^{-n}\sum_x (-1)^{f(x)}$

Problem Set

1. Write $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ in Dirac form. Find its eigenvectors. Where are they on the Bloch sphere?
2. Prove $HXH = Z$ using matrices, then using outer products. Which method is shorter?
3. Starting from $|01\rangle$, $|10\rangle$ and $|11\rangle$ instead of $|00\rangle$, run the Bell circuit. Show you get $|\Psi^+\rangle$, $|\Phi^-\rangle$ and $|\Psi^-\rangle$.
4. Prove $|\Psi^-\rangle$ is entangled. Is $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ entangled?
5. Write U_f for $f(x) = x$ as a 4×4 matrix and check it equals CNOT.
6. Run Deutsch–Jozsa by hand for $n = 2$, $f(x_1x_2) = x_1$. Which z do you always measure?
7. Replace the helper's $|-\rangle$ by $|+\rangle$. Show that $U_f|x\rangle|+\rangle = |x\rangle|+\rangle$, and explain why the algorithm then reads 0 for every f.
8. Three wires $q_2q_1q_0$ (top first). Write $\mathrm{CNOT}(q_2, q_0)$ and $\mathrm{CNOT}(q_0, q_2)$ as sums of $P/I/X$ tensors and as 8×8 matrices. Which basis states does each one swap?

What's Next

Tomorrow — Gates & Circuit Design [1/4]

The full single-qubit gate set (S, T, rotations), controlled-U, measuring in rotated bases, Classical gates, and circuit identities, all in today's notation.

Reading

Nielsen & Chuang

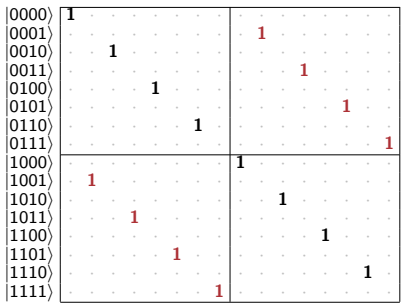
- §2.1: linear algebra in Dirac notation (§2.1.4 outer products, §2.1.7 tensor products)
- §2.2: the postulates
- §1.4.3–1.4.4: Deutsch

Mermin

- Ch. 1: Cbits, Qbits, and operations on them
- §2.2: Deutsch's problem

Now that you have the notation, reread last week's chapters. They will read much faster.

Backup — CNOT(q_0, q_3) in Full



Rows labelled $|q_3 q_2 q_1 q_0\rangle$; columns in the same order.

Black 1s ($q_0 = 0$): on the diagonal, nothing happens.

Red 1s ($q_0 = 1$): flip q_3, i.e. swap

$$|0x_2x_1 1\rangle \leftrightarrow |1x_2x_1 1\rangle$$

- 16 nonzero entries out of 256
- 20 qubits: 10^{12} entries

Nobody writes this out. Simulators store the recipe, not the matrix.

Back